JUNE · IAS CLASS 11

The Talk, Round One

Round Zero was the skeleton. Now there's a manuscript — so today we build the real ten minutes.

IAS 2026 · Yonsei · Saturday 14 November · Free-paper oral · 10 min + 2 min Q&A

THE SETTING

What a Free-Paper Slot Actually Is

IAS10 규정 기준 — IAS11 도 같은 형식일 가능성이 높음

75-minute session, six speakers 10 분 발표 + 2 분 질의 . 좌장이 시간을 끊는다 .

Audience: half clinicians, half researchers PT · AT · 의사 · 연구자 — 심사 기준에 'practical implications' 명시

Slogan: Science ↔ Impact ↔ Practice 임상 함의가 없으면 반쪽 발표

Your track Improving foot-ankle treatment & intervention

Home ground 연세대 주최 · 송경택 교수가 호스트 — 랩 대표 발표

WHAT TEN MINUTES MEANS

- About 1,200–1,400 words in English
- 8–10 slides — one idea per slide
- 30% of the time on results
- Every transition is a time check
- Q&A answers: 30 seconds each, no longer
- If the chair says 'two minutes' — go to the take-home

PART 1 · 0–10 MIN

The One Sentence

청중이 일주일 뒤 기억할 한 문장 . 셋 중 고르거나 섞기 .

A · SURPRISE

Training the foot — not the trunk — changed how the trunk moved during landing.

가장 기억에 남음 . 위험 : ' 왜 ?' 가 바로 나옴 → 기전 답변 필요

B · PRACTICE-FIRST

Four weeks of foot core training is enough to reduce trunk rotation at landing in CAI — and it fits into any protocol.

심사 기준에 정면 . 위험 : 균형 지표 무변화를 앞에서 처리 안 하면 과장

C · HONEST

Neither program improved balance scores. But foot core training changed the movement strategy.

연구자 신뢰 획득 . 위험 : 첫 문장이 부정문이라 힘이 약함

↳ **추천 : A 로 열고 , C 로 결과를 말하고 , B 로 닫는다 . June — which one, and why? Say it in English.**

PART 2 · 10–25 MIN

Three Ways to Open

각각 소리 내서 . 가장 자연스러운 것 하나 .

① STATISTIC

Up to 40% of people who sprain their ankle once go on to develop chronic ankle instability. Most of them never get back to landing the way they used to.

② QUESTION

When you rehab an unstable ankle, where do you start — at the foot, or at the trunk? We asked that question with a four-week trial.

③ SCENE

Picture a volleyball player coming down from a block. The ankle rolls, the knee dips, the trunk twists. Everything we saw in that one landing is what this study is about.

↳ ② 가 가장 안전 — 임상가든 연구자든 자기 답을 떠올리게 만든다 .

THE SCRIPT · S1–S3

Why This Question, and the Aim

0:00 – 2:15 · 약 260 단어

[S1] Good morning. My name is Jiyoon Lee, from the Department of Physical Education at Yonsei University. Today I'd like to share a four-week training study on individuals with chronic ankle instability — and a result that surprised us.

[S2] Up to 40 percent of people who sprain their ankle once go on to develop chronic ankle instability. The main complaint is not pain. It's instability — the ankle gives way, especially during dynamic tasks like jump landing. So when you rehab an unstable ankle, where do you start? At the foot, or at the trunk? Both have a case. At the distal end, people with CAI show weaker intrinsic foot muscles — the so-called foot core. At the proximal end, they show reduced activation of deep trunk stabilizers like the transversus abdominis. Both foot core training and trunk core training have been shown to improve balance in this population. But here's the gap. We don't know whether those improvements carry over to landing — and we've never compared the two approaches head to head on a dynamic task.

[S3] So our aim was simple. We compared foot core training, trunk core training, and a no-training control on three things: static balance during single-leg stance, dynamic balance after a single-leg landing, and lower-limb and trunk kinematics during that landing.

↳ OTHER WAYS TO SAY IT

Bottom-up vs top-down:

"These are two different philosophies — a bottom-up approach starting at the foot, and a top-down approach starting at the trunk."

— 두 훈련의 개념 차이를 한 문장으로

The gap, shorter:

"Both work for balance. Nobody has asked whether they work for landing."

— 시간 부족하면 이 두 문장으로 축약

Aim, shorter:

"We put the two head to head — on balance and on landing."

THE SCRIPT · S4–S5

Methods

2:15 – 4:15 · 약 330 단어

[S4] This was a randomized three-group pretest–posttest design. We recruited thirty participants who met the International Ankle Consortium criteria for CAI — at least one significant sprain more than a year ago, at least two episodes of giving way in the past six months, and cutoff scores on the IdFAI and the FAAM. They were stratified by sex and randomized into three groups of ten: a foot core group, a trunk core group, and a control group. The two training groups completed twelve supervised sessions — thirty minutes, three times a week, for four weeks. The control group received no intervention. [훈련 내용 한 문장 — June 확인 후 삽입]

[S5] We tested before and after the four weeks. For static balance, participants stood on the involved limb for ten seconds, eyes open and eyes closed. We measured center-of-pressure velocity and 95-percent ellipse area. For the dynamic task, they jumped forward over a 30-centimeter hurdle from a distance of 40 percent of their height and landed on the involved limb. From that landing we took two things: the dynamic postural stability index from the force plate, and three-dimensional kinematics of the ankle, knee, hip and trunk from a 35-marker Vicon setup. For the kinematics we looked at the angle at initial contact and the peak within the first 200 milliseconds — and we also ran statistical parametric mapping across that whole window, so we're looking at the entire curve, not just two time points.

↳ OTHER WAYS TO SAY IT

Why SPM, in one line:

"SPM lets us test the whole curve, not just two snapshots."

Numbers spoken aloud:

thirty participants · ten per group · twelve sessions · thirty minutes · three times a week · a thirty-centimetre hurdle · forty percent of body height · two hundred milliseconds

— 숫자는 전부 단어로 연습

Hurdle task, shorter:

"A forward jump over a hurdle, landing on one leg, holding it for ten seconds."

THE SCRIPT · S6–S7

Results

4:15 – 6:45 · 약 330 단어 · 발표의 30%

[S6] First, the balance measures. And here I'll be direct: nothing changed. For every static and dynamic postural control outcome, there was no time-by-group interaction. Four weeks of either program did not move the balance scores more than doing nothing. That was not what we expected. Previous work had shown balance improvements from both approaches. We'll come back to why.

[S7] Now, the kinematics. At the ankle, knee and hip — again, no interaction. But at the trunk, in the transverse plane — that's trunk rotation — we found a significant interaction, both at initial contact and at the peak within 200 milliseconds. And the direction was clear. The foot core group reduced trunk rotation more than the trunk core group. At initial contact, the difference in change scores was significant, with a confidence interval from minus 12 to about minus 1 degree. At peak, the difference was larger — minus 14.6 to minus 2.5 degrees. Foot core versus control just missed significance at peak — p equals 0.053 — and trunk core was no different from control. The SPM analysis tells the same story across the whole curve: the interaction was significant over the entire 200-millisecond window, and the foot core group showed greater reduction than the trunk core group from 25 percent of the window onward, and than control in the final 20 percent. So: we trained the foot, and the trunk changed.

↳ OTHER WAYS TO SAY IT

Three tones for the null result:

Neutral — "There were no significant time-by-group interactions for any postural control outcome."

Direct — "Here I'll be direct: nothing changed."

Contrast — "Balance stayed where it was. Movement didn't."

Headline, three versions:

"Trunk rotation decreased more after foot core than after trunk core training."

"The foot core group showed the greatest reduction in trunk rotation."

"We trained the foot, and the trunk changed."

Saying the CI:

minus twelve point one to minus one point one degrees

THE SCRIPT · S8–S9

What It Means, and the Take-Home

6:45 – 9:45 · 약 360 단어

[S8] How do we make sense of this? One way to think about it is that the foot is the first link in the chain. During landing, the foot is the first thing to hit the ground, and the trunk is the last segment to settle. If the foot handles initial contact better — if it stiffens and stabilizes faster — the body has less to correct upstream. A quieter trunk may be a sign of a quieter foot. The trunk core group trained the segment we measured, and it didn't change. That suggests trunk stabilization in isolation may not transfer to a fast, single-leg landing. Now, the balance scores. Why didn't they change? We think four weeks is enough to shift a movement strategy but not enough to shift a summary measure like COP velocity. And people with CAI compensate — they can hold a stance, but the way they get there is different. Kinematics picked that up; balance scores didn't. Limitations: thirty participants, ten per group. Four weeks, no follow-up. We don't know whether the change lasts, or whether it reduces giving-way episodes.

[S9] So, the take-home for practice. Four weeks of foot core training — twelve short supervised sessions — reduced trunk rotation during landing in people with CAI. Trunk core training, over the same period, did not. If you're building a CAI rehab program, this is an argument for adding a foot core component early — not instead of balance and strength work, but alongside it. It's cheap, it's low-load, and it targets the first link in the chain. Thank you. I'm happy to take questions.

↳ OTHER WAYS TO SAY IT

Mechanism, two metaphors:

Chain — "Fix the first link, and there's less to correct upstream."

Information — "Better foot control means better sensory information reaching the trunk earlier. The trunk doesn't have to guess."

Limitations, short and early:

"Before the take-home, three limitations: small groups, four weeks, no follow-up."

Closing, two ways:

Recommend — "an argument for adding a foot core component early."

Return the question — "So, foot or trunk? For landing mechanics, our data say: start at the foot."

— 이 구간은 원고 Discussion 이 비어 있어 Brian 구성 — June 해석과 대조

WORDS THAT MUST BE RIGHT

Terms, Results Language, Transitions

발음까지 — 좌장 · 청중이 이 단어로 질문한다

TERMS

chronic ankle instability · giving way

intrinsic foot muscles · foot core

trunk core · transversus abdominis

single-leg stance / jump landing

initial contact (IC)

COP velocity · 95% ellipse area

dynamic postural stability index

transverse plane · trunk rotation

statistical parametric mapping

time-by-group interaction

change score · stratified by sex

bottom-up / top-down · distal / proximal

SAYING RESULTS

There was a significant time-by-group interaction for ~

No interaction was found for ~

~ did not differ between groups

The foot core group showed a greater reduction in ~ than ~

This just missed significance (p = 0.053)

The confidence interval ran from ~ to ~ degrees

across the entire 200-millisecond window

from 25 percent of the window onward

TRANSITIONS · HEDGING · AVOID

So our aim was simple.

First, the balance measures.

Now, the kinematics.

How do we make sense of this?

So, the take-home for practice.

One way to think about it is ~

This suggests ~ / It's possible that ~

We can't say ~. We can say ~.

× very significant · × I think + result

× as you can see (2 회 이상) · × sorry

PART 4 · 45–55 MIN

Six Questions, Thirty Seconds Each

Brian = 좌장 + 질문자 . 타이머 30 초 .

Q1 Why no change in balance scores?

↳ Two reasons. Four weeks may change a movement strategy but not a summary measure like COP velocity. And people with CAI compensate — they can hold a stance, but how they get there differs. Kinematics caught that; balance scores didn't.

Q2 Is four weeks enough?

↳ For balance, apparently not. For landing mechanics, yes — trunk rotation changed within four weeks. Whether it holds at eight or twelve weeks, we don't know.

Q3 Ten per group — power?

↳ We powered for 30 on a medium effect. The trunk effects were large — partial eta squared 0.26 and 0.34. The significant result isn't a power problem. The null results might be; I hold those more loosely.

Q4 Mechanism — how does the foot change the trunk?

↳ We can't show mechanism from this design. Our working idea: a stiffer, better-controlled foot at initial contact gives the body less to correct upstream, so the trunk rotates less.

Q5 How would you use this in a clinic?

↳ Add a foot core block early — short, supervised, low load. Not a replacement for balance and strength work; the first thing in the session. Twelve sessions of thirty minutes fit any protocol.

Q6 FCG vs control was p = 0.053. Isn't the headline really FCG vs TCG?

↳ Yes — the clean difference is foot core versus trunk core. Versus control, SPM reached significance in the last 20% of the window, but the discrete peak just missed. A consistent trend, not a proven difference.

TODAY, AND WHAT TO BRING NEXT WEEK

The Hour, and the Gaps

THE HOUR

0–10 The one sentence A·B·C — pick, say why in English

10–25 Three openings Say each aloud. Keep one.

25–45 Results only (S6–S7) Numbers out loud · transitions · 'here I'll be direct'

45–55 Six questions 30-second timer. Brian is the chair.

55–60 Gaps for next week → right column

ASK JUNE — MANUSCRIPT v1 HAS HOLES

· Training Intervention 절이 비어 있음 — 발 코어 · 몸통 코어 훈련이 정확히 뭐였나 ? S4 에 한 문장 필요

· Discussion · Conclusion · Practical Implications 비어 있음 — S8·S9 는 초록 + 결과로 구성한 것 . June 해석과 맞는가 ?

· Figure 1(SPM 상호작용)·Figure 2(pairwise) — 발표용 재작도 필요 (축 라벨 크게)

· 세션 · 순서 · 좌장이 정해졌는지

· 슬라이드는 June 의 IAS 템플릿 v5(마스터 레이아웃) 로